Gauss-Newton on φ versus gradient descent on f for globally PŁ functions

Siddhant Gadodia, IIT Delhi

`https://github.com/[USERNAME]/pl-least-squares-check`

Setup. Boumal, Criscitiello and Rebjock (arXiv:2604.07972) show that a smooth, globally PŁ function on $\mathbb{R}^n$ has the form $f = f^* + \|\varphi(x)\|^2$, where $\varphi : \mathbb{R}^n \to \mathbb{R}^k$ is a submersion and k is the codimension of the minimiser set S. I asked what this structure gives algorithmically when φ is known: how does Gauss-Newton on φ compare with gradient descent on f, and how does the gap depend on the conditioning of $D\varphi$?

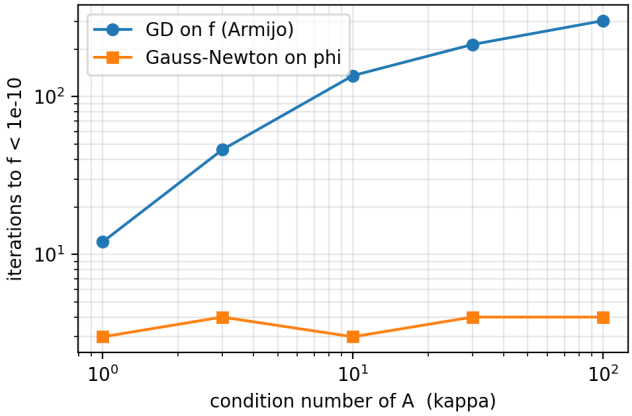
Experiment A (one dimension). $f(x) = x^2 + 3\sin^2 x$ is nonconvex but PŁ. Here $\varphi(x) = \operatorname{sign}(x)\sqrt{f(x)}$. Since $f' = 2\varphi\varphi'$ and PŁ forces $f' \neq 0$ away from 0, φ' cannot vanish there, and $\varphi'(0) = 2$. Numerically on $[-10, 10]$: $\min|\varphi'| \approx 0.296$ and $\min|f'|^2/(2f) \approx 0.176$.

start x_0	-7.3	-2.1	0.9	3.3	8.8
GD on f (step 1/8)	12	7	3	9	13
Gauss-Newton on φ	4	3	3	3	6

iterations to $f < 10^{-10}$

Experiment B (conditioning). $\varphi(x) = Ax + 0.3\sin(Bx)$ from $\mathbb{R}^{20}$ to $\mathbb{R}^5$, with A of condition number κ. Both methods use Armijo backtracking. Gauss-Newton uses the minimum-norm step $-D\varphi^+\varphi$ since $k < n$. Median over five random starts.

κ	GD	Gauss-Newton
1	12	3
3	46	4
10	136	3
30	214	4
100	303	4



Observations. (1) On S, $\nabla^2 f = 2\,D\varphi^\top D\varphi$, so near S Gauss-Newton on φ coincides with Newton on f. The comparison is only informative away from S. (2) Iteration counts understate the cost of Gauss-Newton, which needs a least-squares solve per step. (3) These results are expected once φ is known. The open part, for me, is that φ is usually not available. In the paper it arises from the gradient flow, which raises the question of whether a computable approximation of φ away from S could serve as a preconditioner for first-order methods.